

RP-LEO Platform — Full Technical Documentation

1. Project Overview

Platform Name

RP-LEO Partnership Management Platform

Purpose

The RP-LEO Platform is a full-stack web application designed to manage:

- Partnerships
- Memorandum of Understanding (MoU)
- Document Uploads
- Partner Organizations
- Reports and Monitoring
- User Management

This platform is built for the Space Science and Geospatial Institute (SSGI) Regional Partnership Lead Executive Office (RP-LEO).

2. Tech Stack

Frontend

- React.js
- Tailwind CSS
- Axios / Fetch API
- React Router

Backend

- Node.js
- Express.js
- MongoDB
- Mongoose
- JWT Authentication
- Multer (File Upload)

Development Tools

- VS Code
 - Postman
 - MongoDB Atlas
 - Git & GitHub
-

3. System Architecture

Client (React) ↓ API (Node + Express) ↓ Database (MongoDB) ↓ File Storage (Uploads Folder / Cloud Storage)

4. Project Folder Structure

Root Folder

rp-leo-platform/

client/ server/

Frontend Structure

client/src/

components/ pages/ hooks/ services/ layouts/ utils/

Backend Structure

server/

config/ models/ routes/ middleware/ controllers/ uploads/ server.js

5. Features Overview

Core Features

Authentication System

- Login
 - Register
 - JWT Authentication
 - Role based access
-

Partnership Management

Users can:

- Create partnerships
 - Edit partnerships
 - Delete partnerships
 - View partnerships
-

MoU Management

Users can:

- Create MoU
 - Upload MoU Document
 - Track MoU status
 - Update MoU
-

File Upload System

Supported Files

- PDF
- DOC
- DOCX
- JPG
- PNG

Upload Features

- Drag and Drop Upload
- Progress Bar

- Upload Status
 - File Preview
-

6. Database Models

User Model

Fields

- name
- email
- password
- role
- createdAt

Example

name: String email: String password: String role: admin | staff

Partnership Model

Fields

- organizationName
 - country
 - contactPerson
 - email
 - status
 - createdAt
-

MoU Model

Fields

- title
- partner
- description
- startDate
- endDate
- document
- status
- createdAt

7. API Routes

Authentication

POST /api/auth/register POST /api/auth/login GET /api/auth/me

Partnerships

GET /api/partnership POST /api/partnership PUT /api/partnership/:id DELETE /api/partnership/:id

MoU

GET /api/mou POST /api/mou PUT /api/mou/:id DELETE /api/mou/:id

Upload

POST /api/upload

8. File Upload System

Upload Flow

User Select File ↓ React FormData ↓ POST /api/upload ↓ Multer Middleware ↓ Uploads Folder ↓ Return File URL

9. Frontend Pages

Public Pages

Home About Services Contact

Dashboard Pages

Dashboard Partnerships MoUs Reports Users Settings

10. UI Components

Navbar Sidebar Dashboard Layout Cards Tables Forms Upload Component

11. Authentication Flow

User Login ↓ JWT Token Generated ↓ Token Stored in LocalStorage ↓ Protected Routes

12. Security

JWT Authentication Password Hashing (bcrypt) Protected Routes Role Based Access

13. Environment Variables

server/.env

MONGO_URI= JWT_SECRET= PORT=5000 NODE_ENV=development

14. Installation Guide

Backend

cd server npm install npm run dev

Frontend

cd client npm install npm start

15. Deployment

Frontend

Vercel Netlify

Backend

Render Railway

Database

MongoDB Atlas

16. Future Features

Email Notifications Audit Logs Analytics Dashboard Multi Language Support Mobile App

17. User Roles

Admin

Full Access

Staff

Limited Access

Viewer

Read Only

18. Dashboard Analytics

Total Partnerships Active MoUs Pending Approvals Documents Uploaded

19. File Storage Structure

uploads/

pdf images documents

20. Version

Version 1.0

21. Author

RP-LEO Platform Built using React + Node + MongoDB

End of Documentation